AUtoMLy — Alabama Medicaid Policy Copilot

Milestone 2 status: model selection is converging, protected final evaluation is staged, and the UI/RAG phase is next.

Likely model candidate: NVIDIA Nemotron 3.5 Lightning 30B-A3B LoRA checkpoint 400 — selection is confirmed only by the one-time protected final evaluation.

MILESTONE 1

Complete

Data & model selection; v4 published and resubmission delivered.

MILESTONE 2

In progress

Benchmark tooling passed independent review today; frozen final run is gated, not yet executed.

NEXT PHASE

UI + RAG

Open WebUI retrieval over Medicaid policy; four-way comparison across training and retrieval.

Where We Are in the Project

Three-phase plan: choose data and models, prove the trained model against its base, then build and compare the deployed system.

MILESTONE 1
Complete

Data pipeline and four-family model comparison

Resubmission and v4 delivered; the student-authored 30-question control set is preserved as fixed evidence.

MILESTONE 2
In progress

Nemotron checkpoint-400 vs its untouched base

Identical official-source passages and questions; tooling independently reviewed today; final campaign not yet executed.

UI / RAG PHASE
Next

Open WebUI + ChromaDB over Medicaid policy

Then the four-way comparison: base, trained, base+RAG, trained+RAG — one frozen benchmark, one rubric.

> **Tonight: what is done, why Nemotron is the likely pick, and what comes next.**

The Training Program Is Complete and Clean

Sixteen matched runs — four model families × four data sizes — on fixed cohorts; identical validation IDs; protected rows untouched.

COMPLETED RUNS

16 / 16

All matched cohorts complete as of the Oct 1 clean sweep.

POLICY EXAMPLES

200 – 1,600

Nested train sizes; identical prompt-ID hashes at each size.

PROTECTED ROWS USED

0

No test or student-control rows touched in any run.

N = 1,600 RUN	Best val loss	Best val accuracy	Selected checkpoint
Gemma 4 26B-A4B	0.5750	85.88%	step 400
Muse Glimmer 30B	0.5787	85.54%	step 200
Nemotron 3.5 Lightning 30B-A3B	**0.5527**	**85.56%**	**step 400**
Ornith 1.5 35B-A3B	0.5626	85.38%	step 400

Token loss is a within-family next-token diagnostic — different tokenizers prevent a cross-family accuracy ranking. It motivates the candidate; it does not crown one.

Nemotron Is the Likely Candidate

Three verified reasons — plus one honest boundary on what the current evidence can and cannot claim.

BEST VAL LOSS

0.5527

Lowest best-validation loss across all sixteen matched runs; checkpoint 400 at N=1,600.

CLEAN LOCAL PATH

Verified

All four data sizes completed on lab hardware with rank-16 LoRA after the Oct 1 correction.

BUILT TO SERVE

3B active

30B MoE with 3B active parameters; N=1,600 sweep ran in ~1.4 h vs ~2.5 h for the other MoE family.

Decision status: the Milestone 2 protocol and recorded user decision designate Nemotron N1600 (revision a9904d2, adapter SHA-256 4aaca5c5…) as the candidate.

Formal selection happens only when the sealed, one-time final evaluation passes — no winner from token loss alone.

Evaluation Machinery Under Independent Review

What Milestone 2 has produced so far — reviewed engineering, authenticated sources, and updated program records.

- ✓ **Benchmark authoring tools repaired and passed independent review**
  Answerable-set tool (106 tests) and unanswerable-set tool (44 tests): independent SPEC review, then quality review — both PASS as of today, Oct 15.
- ✓ **Final collection and scoring harness reviewed under synthetic inputs**
  Paired candidate/base collector with sealed-asset checks; passed independent review on synthetic probes only — no real inference run yet.
- ✓ **Fresh benchmark sourced from authenticated official documents**
  12 post-snapshot source snapshots re-hashed; 417 audited page packets; proposed 8-source / 28-page cohort with 6 strict post-publication date alerts.
- ✓ **Strata designed for how policy questions actually fail**
  Policy reasoning · effective dates · exceptions · conflicts · synthesis · unanswerable items that should trigger abstention.
- ✓ **Program records updated**
  Progress Log 2 revised (20 personal hours); Milestone 1 v4 and the Milestone 2 development report published to the public repository.

No final scores exist yet — tonight reports capability built, not model performance.

Paused at the Right Place — Five Gates Remain

Work stops after each gate passes. The pause today was a session-quota boundary, not a blocker in the science.

1	**Integrate reviewed tooling**	Apply the reviewed identity-store upgrade; rerun real-tree tests and the source-only CLI preflight.
2	**Seal the freeze**	Adapter and base bytes, benchmark/key hashes, prompts, settings, metrics, robustness protocol — one immutable manifest.
3	**Run once**	Paired Nemotron-400 vs untouched base on identical official-source passages and questions. One time. Failures kept.
4	**Score honestly**	Machine advisory scoring on the full population; scoped human adjudication stays separate and is not waived.
5	**Review and publish**	Personal rubric review of M2/Log 2; executed notebook; DOCX/PDF; push private and J.O.-only public repos.

Paused at a quota boundary — checkpoint saved; every gate must pass in order.

Open WebUI + RAG: The Four-Way Comparison

The capstone question in one grid: does policy fine-tuning, retrieval, or both actually move answer quality?

Build order

1 Ingest the policy corpus
Administrative Code and Provider Billing Manual, plus audited post-snapshot sources, into ChromaDB.

2 Serve it through Open WebUI
Citation-grounded chat over the Medicaid policy collection for staff use.

3 Deploy both model conditions
Base Nemotron and the selected checkpoint behind one local endpoint.

4 Run the same frozen benchmark
One cohort, one key, one scoring protocol — across four arms.

	BASE MODEL	FINE-TUNED (CKPT 400)
No RAG	What comes out of the box	Value of training alone
With RAG	Value of retrieval alone	The combined hypothesis

Four arms share one frozen benchmark and one human scoring rubric — differences attribute to method, not to the test. Latency and memory are logged per arm.

One sealed benchmark serves all four arms — so the RAG phase follows this evaluation.

Three Things to Take Away

No performance claims tonight — the claims are about readiness, discipline, and what happens next.

1. The experiment loop is real.

Sixteen matched training runs, fixed cohorts, protected rows untouched, and an auditable evidence chain from run records to this slide.

2. Nemotron is the likely choice — and it knows it.

Best validation loss in the sweep, a verified clean training path, and a serving-friendly MoE. The sealed one-time evaluation is the last gate before formal selection.

3. The next build answers the deployment question.

Open WebUI + ChromaDB over Medicaid policy, then base vs trained vs +RAG vs both — one frozen benchmark, one rubric, four arms.

Evidence: github.com/porespellar/INAI6986 (private · full engineering trail) · github.com/porespellar/INAI6986_project_deliverables (public · J.O. deliverables + this deck)